The pump present in this system pumps the water into the water Jacket of the engine quickly. The pumped water takes away the heat from the engine cylinder and goes to the radiator. The hot water in the radiator is cooled as it enters into the shells of the radiator and also due to the opposing air coming from the atmosphere. As the air passes through the gaps of the shells,by touching the fins in the radiator the heat is removed from the water. The cooled water is again sent to the engine water jackets by the pump. By this way the pressurised or pump circulation cooling system rejects heat. The power to the water pump is taken from the V-pulley belt drive.

The important components of this system are as follows,

1. Water jacket
2. Water pump
3. Radiator
4. Thermostat valve
5. Fan and fan belt
6. Radiator hose

The following are the Advantages of the water cooling system:

1. The engine noise is reduced and runs smoothly
2. Cooling rate is high
3. The engine can be operated for more time than other type of cooling system engines
4. This is mostly used for multi cylinder engines
5. Temperature is controlled as per the requirement of the engine

Draw backs:

1. Maintenance is difficult
2. Maintenance cost is higher than air cooled engines
3. Water pump, water jacket, radiator and the radiator fan are the components needed.
4. Purified water has to be used.
5. Deposits and corrosion of the engine water jackets are possible when using impure water in this system

7.3 PARTS OF THE WATER COOLING SYSTEM

1. Radiator
2. Pressure cap
3. Thermostat
4. Water pump
5. Fan

7.3.1 Radiator

Radiator consists of three main parts such as upper tank, core and lower tank. The radiator core and connecting tube are placed in between the upper tank and lower tank. In a radiator the upper tank is connected to the upper surface of the engine through a separate hose. The lower tank is connected with the water pump through a separate hose connection.

The hot water coming out from the engine goes to the upper tank of the radiator through the hose connection. The water reaching the lower tank from upper tank of the radiator is cooled by the radiator core. Before reaching the lower tank the water is cooled by the air passing through the fins of the radiator core by the atmospheric air. The atmospheric air is sucked by the radiator fan and hence the hot water is cooled.

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